



GIRRAWEE HIGH SCHOOL
HALF YEARLY EXAMINATION

YEAR 12

2011

MATHEMATICS
EXTENSION 1

Time allowed – Two hours

(Plus 5 minutes reading time)

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question.
Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Each question attempted is to be returned on a *separate* piece of paper clearly marked Question 1, Question 2, etc. Each piece of paper must show your name.
- You may ask for extra pieces of paper if you need them.

Total Marks – 100

Attempt all questions 1-5

All questions are NOT of equal value.

Answer each question clearly ON A SEPARATE PAGE!

Question 1 (20 Marks) Use a separate piece of paper.		Marks
(a)	Find the ratio in which the point (2,5) divides the interval AB where $A = (4, 9)$, $B = (-3, -5)$.	3
(b)	Expand and simplify $\cos(x - y) - \cos(x + y)$.	2
(c)	Find an expression for $\sqrt{\frac{1 + \cos \theta}{1 - \cos \theta}}$, in terms of t , $t = \tan \frac{\theta}{2}$.	3
(d)	Solve $\frac{2x - 6}{x} < 1$.	4
(e)	If α, β and γ are the roots of the equation $x^3 + 2x^2 + 3x + 4 = 0$, find the value of:	
(i)	$(\alpha - 1)(\beta - 1)(\gamma - 1)$	2
(ii)	$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$	2
(iii)	$\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma}$	2
(f)	Find the coefficient of x^3 in the expansion $(x - 5)^4$.	2

Question 2 (27 Marks) Use a separate piece of paper.

- (a) Find the number of six-letter arrangements that can be made from the letters in the word

SYDNEY.

2

- (b) Differentiate:

(i) $\frac{e^x + 1}{2x}$

3

(ii) $\ln\left(\frac{x}{x^2 + 1}\right)$

3

- (c) Evaluate:

(i) $\int_0^1 x^2 e^{-x^3} dx$

3

(ii) $\int_0^1 2^x dx$

3

- (d) The gradient of a curve is given by $y' = e^{2-x}$ and the curve passes through the point (0,1). Find the equation of the curve and its horizontal asymptote.

5

- (e) (i) Find $\frac{d}{dx}(e^x + e^{-x})$.

1

(ii) Hence, find $\int_0^2 \left(\frac{e^x - e^{-x}}{e^x + e^{-x}} \right) dx$

3

- (f) Find the equation of the normal to $y = e^{-x}$ at the point $P(-1, e)$.

4

Question 3 (17 Marks) Use a separate piece of paper.

- (a) When Amy crossed a tall strain of pea with a dwarf strain of pea, she found that $\frac{3}{4}$ of the offspring were tall and $\frac{1}{4}$ were dwarf.

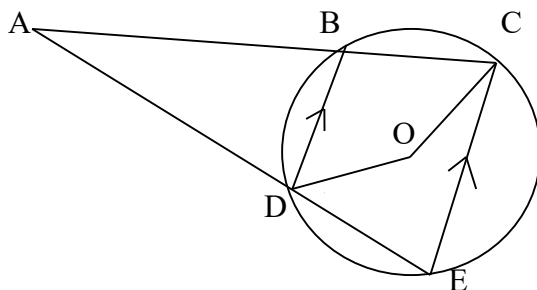
Suppose five such offspring were selected at random.
Find the probability that:

- (i) All of these offspring were tall. 2
- (ii) At least three of these offspring were tall. 3
- (b) Solve the equation $2x^3 - 7x^2 - 12x + 45 = 0$ given that two of its roots are equal. 4
- (c) Given that $(x - 3)$ and $(x + 2)$ are factors of $x^3 - 6x^2 + px + q$, find the values of p and q . 4
- (d) Prove by mathematical induction that for $n \geq 1$
- $$1^2 + 3^2 + \dots + (2n - 1)^2 = \frac{1}{3}n(2n - 1)(2n + 1). \quad \text{4}$$

Question 4 (11 Marks) Use a separate piece of paper.

- (a) Find the acute angle between the lines $x - 2y = 6$ and $4x - y = 1$. 3
- (b) Prove that $\frac{1 - \cos 2\theta}{\sin 2\theta} = \tan \theta$. 2

(c)



BD and CE are two parallel chords of a circle with centre O.
CB and ED produced meet at A. Prove that:

- (i) $AC = AE$ 3
- (ii) ACOD is a cyclic quadrilateral. 3

Question 5 (25 Marks) Use a separate piece of paper.

(a) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$.

(i) If PQ passes through the point $R(2a, 3a)$ show that $pq = p + q - 3$.

5

(ii) If M is the midpoint of PQ , show that the coordinates of M are:

$$\left[a(pq + 3), \frac{a}{2}(pq + 3)^2 - 2pq \right]$$

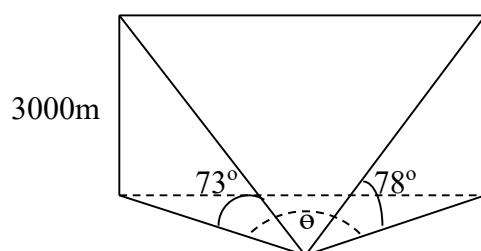
4

(iii) Hence, find the locus of M .

3

(b) A hot air balloon flying at 950m/h at a constant altitude of 3000m is observed to have an angle of elevation of 78° . After 20 minutes, the angle of elevation is 73° . Calculate the angle through which the observer has turned during those 20 minutes.

4



(c) Write $\sqrt{3} \sin x - \cos x$ in the form, $r \sin(x - \alpha)$.

4

(d) Solve using the t-formula $4 \sin \theta - 3 \cos \theta = 2$ for $0^\circ \leq \theta \leq 360^\circ$.

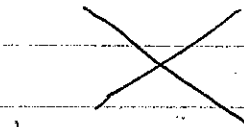
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END OF PAPER.

4R12 HY EXT1 2011.

Question 1.

a) A B
(4, 9) (-3, -5).



k = 1 P(2, 5)

$$2 = \frac{1(4) + k(-3)}{1+k}$$

$$2 + 2k = 4 - 3k$$

$$5k = 2$$

$$k = \frac{2}{5}$$

b) $\cos(x-y) - \cos(x+y)$
 $= [\cos x \cos y + \sin x \sin y] - [\cos x \cos y - \sin x \sin y]$
 $= \cos x \cos y + \sin x \sin y - \cos x \cos y + \sin x \sin y$
 $= 2 \sin x \sin y$

c) $\sqrt{\frac{1 + \cos \theta}{1 - \cos \theta}}$

$$= \sqrt{\frac{1 + \left(\frac{1-t^2}{1+t^2}\right)}{1 - \left(\frac{1-t^2}{1+t^2}\right)}}$$

$$= \sqrt{\frac{1+t^2+1-t^2}{1+t^2} \cdot \frac{1+t^2}{1+t^2-1+t^2}}$$

$$= \sqrt{\frac{2}{2t^2}}$$

$$= \sqrt{\frac{1}{t^2}} = \frac{1}{t}$$

d) By critical points.

$$\frac{2x-6}{x} < 1$$

Equality

$$\frac{2x-6}{x} = 1$$

$$2x-6=x$$

$$x=6$$

Testing.

$$\begin{array}{ccc} \text{xxx} \text{ (x)} & \text{---} & \text{(x)} \text{ xxx} \end{array}$$

$$x=-10 \quad x=1 \quad 6 \quad x=7$$

$$\frac{-2-6}{-1}$$

$$\frac{2-6}{1}$$

$$\frac{14-6}{7}$$

$$4 \nless 1$$

$$-4 < 1$$

$$\frac{8}{7} < 1$$

False

True

False

$$\therefore 0 < x < 6$$

$$\text{OR } \frac{2x-6}{x} < 1$$

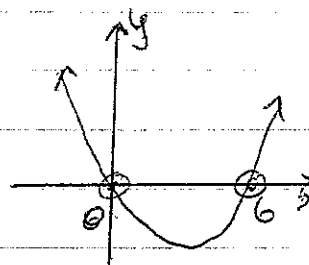
$$\frac{x^2(2x-6)}{x} < x^2$$

$$2x^2-6x < x^2$$

$$x^2-6x < 0$$

$$x(x-6) < 0$$

$$\therefore 0 < x < 6$$



1 cont.

$$\Rightarrow x^3 + 2x^2 + 3x + 4 = 0.$$

$$\alpha + \beta + \gamma = -2.$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = 3$$

$$\alpha\beta\gamma = -4.$$

$$\Rightarrow (\alpha-1)(\beta-1)(\gamma-1)$$

$$= \alpha\beta\gamma - \alpha\beta - \alpha\gamma + \alpha - \beta\gamma + \beta + \gamma - 1$$

$$= \alpha\beta\gamma - (\alpha\beta + \alpha\gamma + \beta\gamma) + \alpha + \beta + \gamma - 1$$

$$= -4 - 3 - 2 - 1.$$

$$= -10.$$

$$\Rightarrow \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$$

$$= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$$

$$= -\frac{3}{4}$$

$$\text{ii)} \frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma}$$

$$= \frac{\gamma + \beta + \alpha}{\alpha\beta\gamma}$$

$$= \frac{1}{2}.$$

$$\text{f)} (x-5)^4$$

$$= x^4 + 4x^3(-5) + 6x^2(-5)^2 + 4x(-5)^3 + (-5)^4$$

$$= x^4 - 20x^3 + 150x^2 - 600x + 625$$

$\therefore$ The coefficient of x^3 is -20 .

Question 2

a) No. of 6 letter arrangements

$$= \frac{6!}{2!}$$

$$= \frac{1 \times 2 \times 3 \times 4 \times 5 \times 6}{1 \times 2}$$

$$= 360.$$

$$\text{bi)} \frac{d}{dx} \frac{e^x + 1}{2x}$$

$$\text{let } u = e^x + 1.$$

$$u' = e^x.$$

$$\frac{dy}{dx} = \frac{2x(e^x) - (e^x + 1)2}{(2x)^2}.$$

$$v = 2x.$$

$$v' = 2$$

$$= \frac{2xe^x - 2e^x - 2}{4x^2}.$$

$$= \frac{2(xe^x - e^x - 1)}{4x^2}.$$

$$= \frac{xe^x - e^x - 1}{2x^2}$$

$$\text{i)} \frac{d}{dx} \ln \left(\frac{x}{x^2 + 1} \right)$$

$$= \frac{d}{dx} \ln x - \ln(x^2 + 1).$$

$$= \frac{1}{x} - \frac{2x}{x^2 + 1}.$$

$$= \frac{(x^2 + 1) - 2x^2}{x(x^2 + 1)}$$

$$= \frac{1 - x^2}{x(x^2 + 1)}$$

Q2 cont.

$$c) \int_0^1 x^2 e^{-x^3} dx.$$

$$= -\frac{1}{3} \int_0^1 -3x^2 e^{-x^3} dx.$$

$$= -\frac{1}{3} \left[e^{-x^3} \right]_0^1.$$

$$= -\frac{1}{3} (e^{-1} - e^0)$$

$$= \frac{1}{3} - \frac{1}{3e}$$

$$= \frac{e-1}{3e}.$$

$$ii) \int_0^1 \frac{x}{2} dx = \int_0^1 e^{x \log 2} dx.$$

$$= \left[\frac{1}{\log 2} e^{x \log 2} \right]_0^1.$$

$$= \left[\frac{2^x}{\log 2} \right]_0^1$$

$$= \left[\frac{2}{\log 2} - \frac{1}{\log 2} \right] = \frac{1}{\log 2}.$$

$$d) y' = e^{2-x}$$

$$\int e^{2-x} dx$$

$$= -\int -e^{2-x} dx$$

$$y = -e^{2-x} + C.$$

passes through (0,1).

$$\therefore 1 = -e^2 + C.$$

$$C = 1 + e^2.$$

$$\therefore y = -e^{2-x} + 1 + e^2.$$

Horizontal Asymptote

$$y = 1 + e^2.$$

$$e) i) \frac{d}{dx} e^x + e^{-x}$$

$$= e^x - e^{-x}$$

$$ii) \int_0^2 \frac{e^x - e^{-x}}{e^x + e^{-x}} dx.$$

$$= \left[\ln(e^x + e^{-x}) \right]_0^2.$$

$$= \ln(e^2 + e^{-2}) - \ln(e^0 + e^0)$$

$$= \ln\left(\frac{e^2 + e^{-2}}{2}\right)$$

Q2 cont.

f) $y = e^{-x}$

$y' = -e^{-x}$ at $P(-1, e)$

$m_{\text{tangent}} = -e$

$m_{\text{normal}} = \frac{1}{e}$

$\therefore \text{Eqn}_{\text{normal}}: y - e = \frac{1}{e}(x + 1)$

$$y - e = \frac{x}{e} + \frac{1}{e}$$

$$ey - e^2 = x + 1$$

$$\therefore x - ey + e^2 + 1 = 0$$

Question 3

a) $t = \frac{3}{4}$ $d = \frac{1}{4}$

i) $P(\text{all tall}) = {}^5C_5 \left(\frac{3}{4}\right)^5$
 $= \frac{243}{1024}$

ii) $P(\text{at least 3 tall})$
 $= {}^5C_3 \left(\frac{3}{4}\right)^3 \left(\frac{1}{4}\right)^2 + {}^5C_4 \left(\frac{3}{4}\right)^4 \left(\frac{1}{4}\right)$
 $+ {}^5C_5 \left(\frac{3}{4}\right)^5$
 $= 10 \left(\frac{27}{4^5}\right) + 5 \left(\frac{3^4}{4^5}\right) + \left(\frac{3^5}{4^5}\right)$
 $= \frac{34(3^3)}{4^5}$

b) $2x^3 - 7x^2 - 12x + 45 = 0$

Roots α, α, β .

$$\alpha + \alpha + \beta = \frac{7}{2}$$

$$2\alpha + \beta = \frac{7}{2} \quad (1)$$

$$\alpha\alpha\beta = -\frac{45}{2}$$

$$\alpha^2\beta = -\frac{45}{2} \quad (2)$$

$$\alpha^2 + 2\alpha\beta = -6 \quad (3)$$

from (1) $\beta = \frac{7}{2} - 2\alpha$

Sub into (2)

$$\alpha^2 + 2\alpha \left(\frac{7}{2} - 2\alpha\right) = -6$$

$$\alpha^2 + 7\alpha - 4\alpha^2 = -6$$

$$-3\alpha^2 + 7\alpha = -6$$

$$3\alpha^2 - 7\alpha - 6 = 0$$

$$(3\alpha + 2)(\alpha - 3) = 0$$

$$\therefore \alpha = 3 \text{ or } -\frac{2}{3}$$

When $\alpha = 3$ $\beta = -\frac{5}{2}$

$$\alpha = -\frac{2}{3} \quad \beta = \frac{29}{6}$$

Sub into (2)

$$\alpha = -\frac{2}{3} \text{ and } \beta = \frac{29}{6} \text{ not a solution.}$$

$$\therefore \alpha = 3 \quad \beta = -\frac{5}{2}$$

Q3 cont.

$$c) x^3 - 6x^2 + px + q$$

$$(x-3)(x+2)$$

$$P(3) = 0$$

$$P(-2) = 0.$$

$$P(3) = 3^3 - 6(3)^2 + p(3) + q$$
$$= -27 + 3p + q$$

$$\therefore 3p + q - 27 = 0. \quad (1)$$

$$P(-2) = (-2)^3 - 6(-2)^2 + p(-2) + q$$
$$= -32 - 2p + q$$

$$\therefore -2p + q - 32 = 0 \quad (2)$$

Subtract (2) from (1)

$$5p + 5 = 0$$

$$5p = -5$$

$$p = -1.$$

$$\therefore 3(-1) + q - 27 = 0.$$

$$-3 + q = 0.$$

$$q = 30.$$

$$\therefore P(x) = x^3 - 6x^2 - x + 30.$$

1) Prove

$$1^2 + 3^2 + \dots + (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1)$$

Prove true for $n=1$.

$$LHS = 1^2 = 1.$$

$$RHS = \frac{1}{3}(1)(2-1)(2+1)$$

$$= 1.$$

$\therefore$ True for $n=1$.

Assume true for $n=k$

$$\therefore 1^2 + 3^2 + \dots + (2k-1)^2 = \frac{1}{3}k(2k-1)(2k+1)$$

Prove true for $n=k+1$

$$1^2 + 3^2 + \dots + (2k-1)^2 + (2k+1)^2 = \frac{1}{3}(k+1)(2k+1)(2k+3)$$

$$LHS = 1^2 + 3^2 + \dots + (2k-1)^2 + (2k+1)^2$$

$$= \frac{1}{3}k(2k-1)(2k+1) + (2k+1)^2.$$

$$= (2k+1) \left[\frac{1}{3}k(2k-1) + (2k+1) \right]$$

$$= (2k+1) \left[\frac{k(2k-1) + 3(2k+1)}{3} \right]$$

$$= \frac{1}{3}(2k+1)(2k^2 - k + 6k + 3)$$

$$= \frac{1}{3}(2k+1)(2k^2 + 5k + 3)$$

$$= \frac{1}{3}(2k+1)(2k+3)(k+1)$$

$$= \frac{1}{3}(k+1)(2k+1)(2k+3)$$

$$= RHS$$

$\therefore$ If statement true for $n=k$,
it is also true for $n=k+1$.

$\therefore$ By the principal of mathematical
induction true for all $n \geq 1$.

Question 4

$$\begin{aligned} \text{a) } x - 2y &= 6 & 4x - y &= 1 \\ y &= \frac{x}{2} - 3 & y &= 4x - 1 \end{aligned}$$

$$\begin{aligned} \tan \theta &= \left| \frac{\frac{1}{2} - 4}{1 + (\frac{1}{2})(4)} \right| \\ &= \left| \frac{-\frac{7}{2}}{3} \right| \\ &= \left| -\frac{7}{6} \right| = \frac{7}{6} \end{aligned}$$

$$\therefore \tan \theta = \frac{7}{6}$$

$$\therefore \theta = 49^\circ 24'$$

$$\text{b) } \frac{1 - \cos 2\theta}{\sin 2\theta} = \tan \theta$$

$$\text{LHS} = \frac{1 - \cos 2\theta}{\sin 2\theta}$$

$$= \frac{1 - (1 - 2\sin^2 \theta)}{\sin 2\theta}$$

$$= \frac{2\sin^2 \theta}{\sin 2\theta}$$

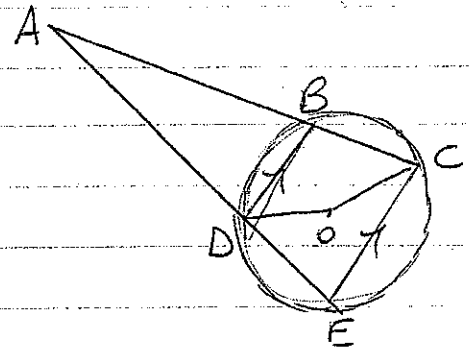
$$= \frac{2\sin^2 \theta}{2\sin \theta \cos \theta}$$

$$= \frac{\sin \theta}{\cos \theta}$$

$$= \tan \theta$$

$$= \text{RHS.}$$

c)



i) Prove $AC = AE$.

$\angle ABD = \angle BCE$ (corresponding $\angle$, $BD \parallel CE$)

$\angle ABD = \angle CED$ (exterior $\angle$ of a cyclic quadrilateral)

$\therefore \angle BCE = \angle CED$

$\therefore \triangle ACE$ is an isosceles $\triangle$

$\therefore AC = AE$ (sides opposite equal angles in isosceles $\triangle$ are equal)

iii) Prove $ACOD$ is cyclic.

$\angle DOC = 2 \times \angle DEC$ (angles on the same arc subtended at the circumference)

$\therefore \angle DOC = \angle DEC + \angle ECB$

(angles in isosceles $\triangle$ from (i))

$\therefore \angle DEC + \angle ECB + \angle CAD = 180^\circ$

(angle sum of a $\triangle$)

$\therefore \angle DOC + \angle CAD = 180^\circ$

$\therefore ACOD$ is a cyclic quadrilateral (opposite angles are supplementary)

Question 5.

a) $P(2ap, ap^2)$ $Q(2aq, aq^2)$
 $x^2 = 4ay$.

i) $m_{PQ} = \frac{ap^2 - aq^2}{2ap - 2aq}$

$$= \frac{a(p^2 - q^2)}{2a(p - q)}$$

$$= \frac{p+q}{2}$$

Eqn_{PQ}: $y - ap^2 = \left(\frac{p+q}{2}\right)(x - 2ap)$

$$y - ap^2 = \left(\frac{p+q}{2}\right)x - 2ap\left(\frac{p+q}{2}\right)$$

$$y - ap^2 = \left(\frac{p+q}{2}\right)x - ap^2 - apq$$

$$y = \left(\frac{p+q}{2}\right)x - apq$$

When $R(2a, 3a)$.

$$3a = \left(\frac{p+q}{2}\right)2a - apq$$

$$3a = a(p+q) - apq$$

$$3 = (p+q) - pq$$

$$\therefore pq = p+q - 3.$$

$$p+q = pq + 3. \text{ (rearranged)}$$

ii) $M_{PQ} = \left[\frac{2ap + 2aq}{2}, \frac{ap^2 + aq^2}{2} \right]$

$$= \left[\frac{2a(p+q)}{2}, \frac{a(p^2+q^2)}{2} \right]$$

$$= \left[a(p+q), \frac{a(p^2+q^2)}{2} \right]$$

$$= \left[a(pq+3), \frac{a}{2}(p^2+q^2) \right]$$

$$= \left[a(pq+3), \frac{a}{2}((p+q)^2 - 2pq) \right]$$

$$= \left[a(pq+3), \frac{a}{2}((pq+3)^2 - 2pq) \right]$$

iii) Let $x = a(pq+3) \Rightarrow pq = \frac{x}{a}$.

$$y = \frac{a}{2}(pq+3)^2 - 2pq$$
$$\Rightarrow \frac{2y}{a} + 2pq = (pq+3)^2$$

$$x^2 = a^2(pq+3)^2$$

$$x^2 = a^2\left(2pq + \frac{2y}{a}\right)$$

$$x^2 = a^2\left[\left(\frac{2y}{a}\right) + 2\left(\frac{x}{a} - 3\right)\right]$$

$$x^2 = 2ay + 2ax - 6a^2$$

$$x^2 - 2ax = 2ay - 6a^2$$

$$\therefore x^2 - 2ax = 2a(y - 3a)$$

b) Balloon travels $\frac{950}{3}$ m

$$\frac{1}{\tan 78^\circ} = \frac{x}{3000}$$

$$\therefore x = \frac{3000}{\tan 78^\circ}$$

$$\frac{1}{\tan 73^\circ} = \frac{y}{3000}$$

$$\therefore y = \frac{3000}{\tan 73^\circ}$$

$$\cos \theta = \frac{\left(\frac{3000}{\tan 78^\circ}\right)^2 + \left(\frac{3000}{\tan 73^\circ}\right)^2 - \left(\frac{950}{3}\right)^2}{2\left(\frac{3000}{\tan 78^\circ}\right)\left(\frac{3000}{\tan 73^\circ}\right)}$$

$$\cos \theta = 0.981068 \dots$$

$$\theta = 11.16^\circ$$

$$\therefore \theta = 11^\circ 10'$$

c) $\sqrt{3} \sin x - \cos x = r \sin(x - \alpha)$

$$\sqrt{3} \sin x - \cos x = r(\sin x \cos \alpha - \cos x \sin \alpha)$$

$$= r \cos \alpha \sin x - r \sin \alpha \cos x$$

$$\therefore r \cos \alpha = \sqrt{3} \quad \text{①}$$

$$r \sin \alpha = +1 \quad \text{②}$$

$$r^2(\cos^2 \alpha + \sin^2 \alpha) = 4$$

$$r^2 = 4$$

$$r = 2$$

$$\therefore \cos \alpha = \frac{\sqrt{3}}{2} \text{ and } \sin \alpha = \frac{+1}{2}$$

α is in 1st quad.

$$\therefore \tan \alpha = \frac{1}{\sqrt{3}}$$

$$\therefore \alpha = 30^\circ$$

$$\sqrt{3} \sin x - \cos x = 2 \sin(x - 30^\circ)$$

d) $4 \sin \theta - 3 \cos \theta = 2$

$$4\left(\frac{2t}{1+t^2}\right) - 3\left(\frac{1-t^2}{1+t^2}\right) = 2$$

$$8t - 3 + 3t^2 = 2(1+t^2)$$

$$3t^2 + 8t - 3 = 2 + 2t^2$$

$$t^2 + 8t - 5 = 0$$

$$t = \frac{-8 \pm \sqrt{8^2 - 4(1)(-5)}}{2}$$

$$\therefore t = 0.58 \text{ or } -8.58$$

$$\frac{\theta}{2} = 30^\circ 7', 210^\circ 7', 96^\circ 39', 276^\circ 3'$$

$$\therefore \theta = 60^\circ 14', 193^\circ 18'$$

Check $\theta = 180^\circ$

$$4 \sin 180^\circ - 3 \cos 180^\circ = 2$$

$$0 + 3 = 2$$

$$3 \neq 2$$

$\therefore \theta = 180^\circ$ is not a solution